Question 4.1 Describe a situation or problem from your job, everyday life, current events, etc., for which a clustering model would be appropriate. List some (up to 5) predictors that you might use.

Solution

As someone who uses public transportation on a daily basis, I am usually angry at the management of the buses and trains in my city in terms of frequency of the rides, cost, cleanliness, breakdowns, road closures, etc. . As a thought experiment, using a clustering model to improve services looks promising. Rather than treating all the bus stops the same, clustering the stops on the basis of the key problems would offer richer data points and probably some effective solutions.

Predictors: average weekday riders, average weekend riders , peak-hour rider counts, off-peak hour rider counts, population within 500 mts of the train/bus stops, number of boardings per day.

Question 4.2

The iris data set iris.txt contains 150 data points, each with four predictor variables and one categorical response. The predictors are the width and length of the sepal and petal of flowers and the response is the type of flower. The data is available from the R library datasets and can be accessed with iris once the library is loaded. It is also available at the UCI Machine Learning Repository (https://archive.ics.uci.edu/ml/datasets/Iris). The response values are only given to see how well a specific method performed and should not be used to build the model.

Use the R function kmeans to cluster the points as well as possible. Report the best combination of predictors, your suggested value of k, and how well your best clustering predicts flower type.

Solution

Methodology

The goal of this analysis was to determine whether flowers could be grouped into similar categories based on their physical measurements. Since the flower species labels were not used to create the groups, this is an example of unsupervised learning.

The dataset contained four predictor variables:

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width

The `scale()` function was used to standardize the predictors so that each variable contributed equally to the clustering process.

The K-means clustering algorithm was implemented using the `kmeans()` function in R. Several values of k (the number of clusters) were tested. The model was run with `nstart = 25`, which allows K-means to try multiple random starting points and select the best solution

Model used in R
kmeans(x_scaled, centers = 3, nstart = 25)

where:

- `x_scaled` contained the standardized predictor variables
- `centers = 3` specified three clusters
- `nstart = 25` improved the reliability of the solution

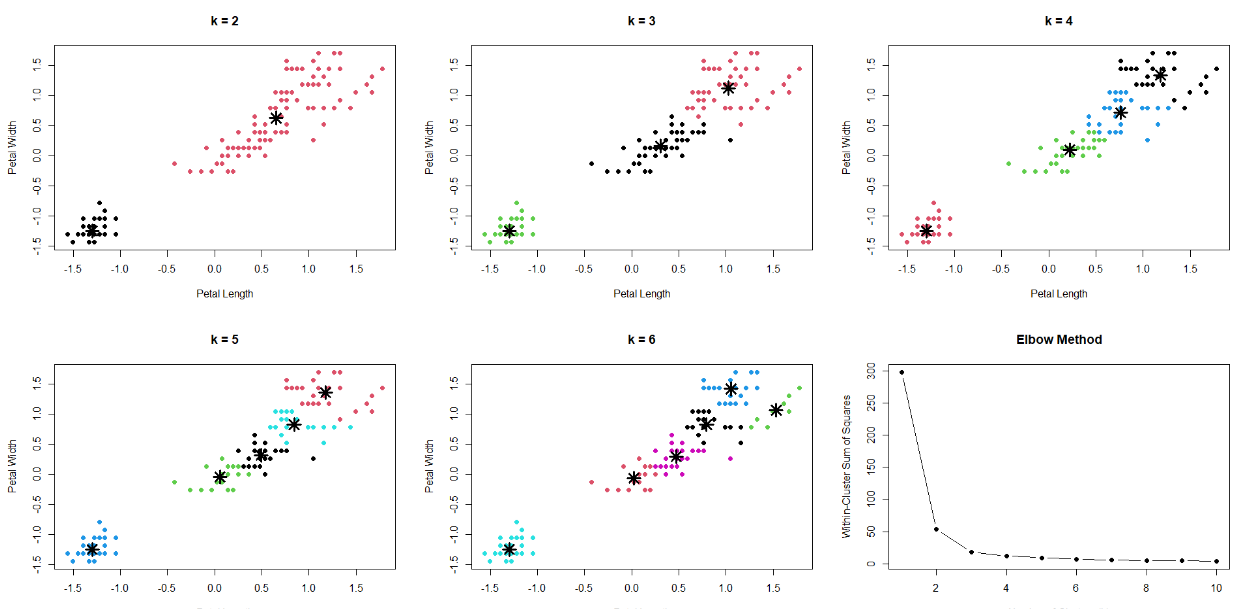
Graphs

The following visualizations were used:

1. Scatter plot of Petal Length vs. Petal Width colored by cluster assignment.
2. Elbow plot showing within-cluster variation for different values of k.

The elbow plot suggested that k = 3 was an appropriate choice because additional clusters provided only small improvements.

Results



Discussion of Results

The results indicate that the Iris dataset contains naturally occurring groups that closely correspond to the three flower species. In particular, petal measurements were highly effective at distinguishing between species.

Question 5.1

Using crime data from the file uscrime.txt (http://www.statsci.org/data/general/uscrime.txt, description at http://www.statsci.org/data/general/uscrime.html), test to see whether there are any outliers in the last column (number of crimes per 100,000 people). Use the grubbs.test function in the outliers package in R.

Methodology

Load the crime data.
Extract the last column (Crime rate per 100,000 people).
Run Grubbs' test to check for an outlier.
Interpret the p-value.

The `grubbs.test()` function from the outliers package was used to determine whether the crime rate variable (number of crimes per 100,000 people) contains an outlier. Grubbs' test compares the most extreme observation to the rest of the dataset and evaluates whether it is unusually far from the mean.

Grubbs' Test is a statistical test used to determine whether the most extreme value in a dataset is an outlier.

Hypotheses

Null Hypothesis (H_0): There are no outliers in the data.

Alternative Hypothesis (H_1): The most extreme observation is an outlier.

For our crime data: Highest crime rate = 1993

Grubbs' Test checked whether 1993 is unusually far from the other crime rates.

Results
G = 2.81287, meaning the value 1993 is about 2.8 standard deviations above the mean.
p-value = 0.07887

Decision rule:

- p-value < 0.05 → Outlier detected
- p-value ≥ 0.05 → No significant outlier

0.07887 > 0.05

There is not enough evidence to classify the crime rate of 1993 as an outlier.

Discussion

Grubbs' test was used to determine whether the highest crime rate in the dataset is statistically different from the remaining observations. The test produced a p-value of 0.0789, which is greater than the common significance level of 0.05. Therefore, we accept the null hypothesis that there are no outliers in the crime rate variable.

Although the observation with a crime rate of 1993 appears unusually high and was identified as the most extreme value, there is insufficient statistical evidence to conclude that it is a true outlier. In other words, the value is high, but not high enough relative to the variation in the rest of the dataset to be considered statistically unusual.

The boxplot of crime rates shows that most observations are concentrated within a relatively narrow range, while a small number of states have substantially higher crime rates. The distribution appears slightly right-skewed, indicating that high crime rates are more spread out than low crime rates. The highest crime rate value (1993 crimes per 100,000 people) appears near the upper end of the distribution and may be displayed as a potential outlier on the boxplot. However, Grubbs' test produced a p-value of 0.0789, suggesting that this observation is not statistically significant as an outlier at the 5% significance level. Therefore, while the boxplot highlights an unusually large value, it should be interpreted as part of the natural variation within the dataset rather than as an extreme anomaly.

6.1

Describe a situation or problem from your job, everyday life, current events, etc., for which a Change Detection model would be appropriate. Applying the CUSUM technique, how would you choose the critical value and the threshold?

Solution

My city transportation dept can track the average delay time for buses on a route each day. Under normal conditions, delays may vary because of traffic, weather, or passenger volume. However, if the average delay suddenly increases and stays high, it could indicate a real change in service performance.

CUSUM would track whether bus delays are consistently above the normal average.

Let:

- X_t = average bus delay on a given day
- μ_0 = normal average delay
- k = critical value
- h = threshold

The CUSUM statistic would increase when delays are repeatedly higher than normal.

Using CUSUM, I would choose the critical value based on the size of delay increase that matters operationally, such as a 4-minute increase. This gives k=2k . I would choose the threshold based on normal delay variation, often around 4 to 5 standard deviations. If the standard deviation is 3 minutes, then h=15. This setup balances early detection of real service problems while reducing false alarms from normal daily variation.

Question 6.2

1. Using July through October daily-high-temperature data for Atlanta for 1996 through 2015, use a CUSUM approach to identify when unofficial summer ends (i.e., when the weather starts cooling off) each year. You can get the data that you need from the file temps.txt or online, for example at http://www.iweathernet.com/atlanta-weather-records or https://www.wunderground.com/history/airport/KFTY/2015/7/1/CustomHistory.html . You can use R if you'd like, but it's straightforward enough that an Excel spreadsheet can easily do the job too.

Methodology

Mean = average daily high temperature
SD = standard deviation
k = 0.5 × SD
h = 5 × SD

CUSUM = min(0, previous CUSUM + (temperature - mean) + k)

CUSUM < -h

Example : 2015

Mean = 83.30
SD = 8.71
k = 0.5 × 8.71 = 4.35
h = 5 × 8.71 = 43.55

Threshold = -43.55

Day	Temp	Temp - Mean	+ k	CUSUM
25-Sep	67	-16.30	-11.95	-19.73
26-Sep	71	-12.30	-7.95	-27.68
27-Sep	71	-12.30	-7.95	-35.62
28-Sep	75	-8.30	-3.95	-39.57
29-Sep	77	-6.30	-1.95	-41.52
30-Sep	85	1.70	6.05	-35.46
1-Oct	71	-12.30	-7.95	-43.41
2-Oct	66	-17.30	-12.95	-56.35

2015 unofficial summer end = 2-Oct

Results for All Years

Year	Mean	SD	k	h	Summer Ends
1996	83.72	8.55	4.27	42.74	2-Oct
1997	81.67	9.32	4.66	46.60	17-Oct
1998	84.26	6.41	3.20	32.05	8-Oct
1999	83.36	9.72	4.86	48.62	7-Oct
2000	84.03	9.52	4.76	47.59	7-Oct
2001	81.55	8.22	4.11	41.12	1-Oct
2002	83.59	9.43	4.71	47.13	15-Oct
2003	81.48	7.02	3.51	35.09	2-Oct

2004	81.76	6.66	3.33	33.31	12-Oct
2005	83.36	7.73	3.87	38.67	13-Oct
2006	83.05	9.79	4.90	48.97	15-Oct
2007	85.40	9.03	4.52	45.17	19-Oct
2008	82.51	8.73	4.37	43.67	20-Oct
2009	80.99	9.01	4.51	45.07	15-Oct
2010	87.21	7.45	3.72	37.23	2-Oct
2011	85.28	9.93	4.97	49.66	10-Oct
2012	84.65	9.25	4.63	46.26	8-Oct
2013	81.67	7.73	3.86	38.63	19-Oct
2014	83.94	6.59	3.30	32.96	29-Sep
2015	83.30	8.71	4.35	43.55	2-Oct

Conclusion: Most years show Atlanta's unofficial summer ending in early-to-mid October.

2. Use a CUSUM approach to make a judgment of whether Atlanta's summer climate has gotten warmer in that time (and if so, when).

Solution

First, the average July–October temperature was calculated for each year, producing one summer temperature value per year. The overall mean and standard deviation of these yearly averages were then computed.

An upper-sided CUSUM was used because the objective was to detect increases in temperature over time. The CUSUM statistic was calculated as:

Ct=max(0,Ct−1+(Xt−μ−k))

where:

- Xt = average summer temperature for year ttt
- μ = overall average summer temperature
- k = critical value

- Ct = cumulative sum statistic

The critical value was chosen as:

k=0.5σ
h=5σ

Results

The average summer temperature across all years was approximately 83.35°F, with a standard deviation of approximately 1.55°F.

K = 0.78 (approx)
h= 7.75 (approx)

Several summers after 2007 were warmer than the long-term average. The CUSUM chart showed a noticeable increase beginning around 2007 and reaching its strongest level around 2010–2012. However, the cumulative sum did not exceed the decision threshold, indicating that the warming signal was not strong enough to be classified as a statistically significant long-term shift.

3. Get your favorite AI to repeat part 2; compare its answer with yours, and judge where each one is better and worse, and (if at all) where each one is wrong. Include the prompt/chat logs in your homework submission, and specify which AI you use

Solution
(I used Copilot , and used the question as a prompt. I used the link to the data to provide the AI access to the data)

The AI asked me to define summer, choose months that define summer. Copilot also commented that it is excellent data.

Here is an abridged version of the response from Copilot

Short answer
Yes — Atlanta's warm season shows a clear upward shift, with the CUSUM indicating a major warming break around 2007–2010.

CUSUM turning points
Minimum CUSUM around 2003 → end of cooler era
Strong positive slope 2007–2012 → structural warming shift
Peak around 2012 → warmest sustained period
2013 dip is noise, not a reversal
2014–2015 return to warm baseline

Conclusion
Atlanta’s warm season shows:
A cool period: 1996–2003
A transition: 2004–2007
A new warmer regime: 2010–2015
The CUSUM identifies 2007–2010 as the most likely change-point.

Final Judgment

Yes — Atlanta’s warm-season climate warmed over 1996–2015.
The CUSUM indicates the warming shift occurred around 2007–2010.
This aligns with broader Southeast U.S. warming patterns after the early-2000s plateau.

Comparing the responses, Copilot is more decisive about the warming period and the analysis of the data. Copilot did not limit its analysis to temps.txt and used the information from the web to justify its response.